ModelIt!

Teacher's Guide

Designing a Mars Habitat That Won't Kill You

Modeling the Interdependent Systems of Extraterrestrial Human Survival

Quick Reference

Grade Level	Grade 10L3
NGSS Standard	HS-ESS1-4, HS-ETS1-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: HS-ESS1-4, HS-ETS1-1

Use mathematical or computational representations to predict the motion of orbiting objects in the solar system. Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students use computational models to predict how energy constraints cascade through interconnected habitat systems and calculate minimum viable operating parameters.
Disciplinary Core Idea	**ESS1.B Earth and the Solar System / ETS1.A Defining and Delimiting Engineering Problems** — Students apply knowledge of Mars' orbital mechanics, atmospheric conditions, and radiation environment to define quantitative criteria for habitat engineering solutions.
Crosscutting Concept	**Systems and System Models** — Students model the Mars habitat as a complex system where feedback loops, energy flows, and interdependencies determine whether the system sustains life or fails catastrophically.

CAST Testing Alignment

MULTIPLE CHOICE:

A Mars habitat's solar panels are producing only 45% of nominal power during a dust storm. According to your model, which system should be reduced LAST to maximize crew survival time?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain why Psychological Well-Being is classified as a survival-critical system rather than a comfort variable. Describe at least two feedback loops between crew mental state and habitat system performance.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Designing a habitat for Mars is the most complex systems engineering challenge humanity has ever attempted. Unlike the International Space Station, which orbits just 400 km above Earth and can be resupplied in hours, a Mars habitat must function autonomously for years with no possibility of quick rescue. Every system depends on every other system, and the margin for error is measured in hours, not days.

The Mars Environment Challenge

Mars presents a hostile combination of threats: atmospheric pressure at 0.6% of Earth's (instant ebullism without a pressure suit), 95% CO_2 atmosphere with no breathable oxygen, average temperatures of -60 degrees Celsius with swings of 100+ degrees daily, radiation doses 200+ times Earth's surface level due to the absent magnetosphere, and global dust storms lasting months that can reduce solar energy by 97%. Every one of these threats is independently lethal.

Systems Interdependence

A Mars habitat is a web of interdependencies. Atmosphere control requires energy for electrolysis and CO_2 scrubbing. Water recycling requires energy and produces oxygen as a byproduct. Food production requires water, light energy, CO_2, and stable temperatures. Waste management feeds composted material back to food production. Temperature regulation demands constant energy. When one system degrades, others follow in a predictable cascade that engineers must anticipate and design for.

The Energy Bottleneck

Energy is the master variable in Mars habitat design. Solar panels on Mars receive only 43% of the sunlight Earth gets, and global dust storms can reduce that to 3% for weeks. Nuclear fission reactors like NASA's Kilopower provide reliable baseline power but are heavy and difficult to maintain. The habitat must balance solar's mass efficiency against nuclear's reliability, with enough total capacity to run all systems plus a safety margin for emergencies.

Human Factors in Extreme Isolation

Psychological research from Antarctic stations, submarine deployments, and ISS missions shows that crew mental health degrades predictably in isolation: performance drops after 3-4 months, interpersonal conflicts intensify, and decision-making quality decreases. On Mars, with 4-24 minute communication delays to Earth and no possibility of evacuation, psychological well-being is not a luxury — it directly affects whether the crew can maintain the systems keeping them alive.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify nine key components of a Mars habitat system: radiation shielding, atmosphere control, water recycling, food production, energy generation, structural integrity, temperature regulation, waste management, and psychological well-being.

E **Establish Relationships**

Students map the complex web of relationships between components, identifying energy generation as the keystone variable and tracing how its reduction cascades through atmosphere, water, food, temperature, and ultimately psychological well-being.

V **Visualize & Model**

Students build a computational model showing the nine-component habitat system with feedback loops, energy dependencies, and cascade failure pathways.

E **Evaluate Outcomes**

Students run nominal operations, dust storm crisis, and water system failure scenarios to test their model and identify critical thresholds for each system.

R **Revise & Extend**

Students add communication relay, medical capability, or spare parts inventory to model even more realistic habitat dynamics and failure modes.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"You have eight crew members, nine critical systems, and a dust storm that just cut your power in half. Everyone's life depends on the decisions you make in the next 30 minutes."*

Watch For

Open with a simulated mission control emergency alert. Let the tension build before revealing the lesson topic.

Slide 2: Learning Objectives

3 min

Say: *"Today you are not just learning about Mars — you are engineering the systems that will determine whether humans live or die 225 million kilometers from help."*

Watch For

Have students read objectives. Pre-teach 'closed-loop life support' and 'systems redundancy.' Emphasize this is about engineering interconnected systems.

Slide 3: Big Question

3 min

Say: *"We know HOW to build each of these systems individually. The hard part is making nine systems work together when failure of any one can kill everyone."*

Watch For

Quick-write: Students list the systems they think are most critical and rank them 1-9. Share out and debate rankings.

Slide 4: LEVER Framework

2 min

Say: *"We are building a model of the most complex engineering system ever designed — a self-sustaining habitat on another planet."*

Watch For

Preview each LEVER step. Emphasize that this model has nine components because real engineering problems are complex and interconnected.

Slide 5: Activity 1: Components

8 min

Say: *"Nine systems. Each one is critical. But some can degrade temporarily while others cannot drop at all. Which is which?"*

Watch For

Guide students through all nine components. Discuss why Energy Generation and Psychological Well-Being are classified as external — they are the inputs that drive the system.

Slide 6: Activity 2: Connections

8 min

Say: *"This is where it gets real. Energy connects to almost everything. When you pull one thread, the whole web moves."*

Watch For

Help students map the dependency web. Use colored arrows to show energy dependencies versus biological dependencies versus psychological feedback loops.

Slide 7: Activity 3: Simulation

10 min

Say: *"Dust storm incoming. You have 40% power for 45 days. Who lives, who dies, and what decisions do you make?"*

Watch For

Students predict cascade sequences before running simulations. Compare predictions to model outputs. Discuss triage decisions.

Slide 8: Discoveries

5 min

Say: *"The difference between a dead crew and a surviving crew is not technology — it is systems design, redundancy, and the ability to degrade gracefully."*

Watch For

Lead discussion connecting model findings to real NASA design philosophy. Discuss why psychological well-being emerged as a critical survival variable.

Slide 9: STEM Challenge

12 min

Say: *"NASA needs your protocol. 45-day dust storm. 40% power. Eight lives. What is your plan?"*

Watch For

Groups design crisis response protocols using model evidence. Present protocols with justification from simulation data. Class votes on most survivable plan.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Energy Generation, Psychological Well-Being
- **INTERNAL:** Radiation Shielding, Atmosphere Control, Water Recycling, Food Production, Structural Integrity, Temperature Regulation, Waste Management

Relationships

- **Energy Generation to Atmosphere Control:** POSITIVE (+) — Higher energy generation powers the electrolysis units that split CO2 into oxygen and the fans that circulate air through scrubbers. When energy drops, atmosphere processing slows and CO2 levels rise dangerously.
- **Water Recycling to Food Production:** POSITIVE (+) — Higher water recycling efficiency means more purified water is available for hydroponic crop irrigation. When water recycling degrades, food production drops because plants cannot grow without consistent water supply.
- **Food Production to Psychological Well-Being:** POSITIVE (+) — Adequate food production provides nutrition and the psychological comfort of fresh food variety. When food production drops and the crew shifts to emergency rations, morale and mental health decline measurably.

Key Discoveries

- Mars habitat systems are so deeply interconnected that degradation in any single system cascades through multiple others within days to weeks
- Energy generation is the master variable — nearly every other system depends on continuous power, making energy redundancy the highest engineering priority
- Psychological well-being is not a luxury but a survival variable — crew mental health directly affects decision-making, maintenance quality, and conflict resolution during crises
- The habitat must be designed for graceful degradation rather than binary success or failure — the ability to operate at reduced capacity keeps the crew alive during emergencies

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Mars habitats just need to be airtight and warm"	Maintaining breathable air and comfortable temperature are necessary but nowhere near sufficient. A sealed, warm box without radiation shielding exposes the crew to 0.67 millisieverts per day — enough to significantly increase cancer risk within months. Without water recycling, the crew runs out of water in weeks. Without food production, stored supplies last 1-2 years at most. Mars habitat design is a nine-dimensional optimization problem, not a two-variable one.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Solar panels work the same on Mars as on Earth"	Mars receives only 43% of the solar energy Earth does because it is 1.5 times farther from the Sun. During global dust storms, which occur every few Mars years and last months, solar energy can drop to as low as 3% of Earth-normal levels. This means a Mars habitat cannot rely solely on solar power the way a house on Earth can — it needs either massive battery storage, nuclear backup, or both.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Psychological well-being is a comfort issue, not an engineering issue"	Research from Antarctic stations, submarines, and the ISS consistently shows that crew psychological deterioration directly causes system failures. The Mars-500 isolation experiment (520 days in a simulated Mars habitat) documented sleep disruption, cognitive decline, interpersonal conflicts, and motivation loss that would have compromised system maintenance on a real mission. On Mars, a crew member who is too depressed to perform maintenance checks could miss a failing component that kills everyone.